Technical Analysis Assistant: A Multi-Agent AI System for Market Pattern Recognition and Trade Analysis

Problem Statement and Motivation

Retail traders and beginner investors often struggle to interpret price charts and technical indicators. While charting platforms provide raw data and visual tools, understanding patterns such as trends, breakouts, support and resistance zones, and momentum requires experience. Many traders misinterpret signals or act without structured risk analysis, leading to inconsistent decision-making.

This project addresses the lack of an intelligent assistant that can interpret data charts, explain technical patterns in plain language, and evaluate potential risks. The target users are beginner and intermediate traders who want structured, explainable insights rather than automated trading decisions.

An agentic AI solution is appropriate because technical analysis is not a single-step process. It involves pattern recognition, contextual reasoning, explanation, and critique. A multi-agent system allows specialized agents to collaborate in stages, improving transparency and reliability compared to a single model making opaque decisions. This architecture mirrors how human analysts separate observation, interpretation, and risk evaluation.

Use Cases and Scenarios

Scenario 1: Pattern Identification

A user uploads historical price data in CSV format. The system analyzes the data and identifies technical patterns such as trends, breakouts, or consolidation zones. The assistant explains what the detected pattern means and why it may be relevant to future price behavior.

Scenario 2: Trade Explanation

A user asks whether a chart suggests bullish or bearish momentum. The system evaluates indicators such as moving averages and volatility, summarizes the technical picture, and explains the interpretation in accessible language suitable for educational purposes.

Scenario 3: Risk Critique

A user proposes a hypothetical trade based on chart signals. The system evaluates the trade idea, highlights uncertainty and risk factors, and suggests alternative interpretations. The goal is to provide learning-focused feedback rather than automated execution.

Initial System Design

The proposed system consists of a structured multi-agent workflow in which each agent has a specialized role.

The Chart Reader Agent processes uploaded price data and extracts features such as trends, support and resistance levels, and indicator values.

The Pattern Recognition Agent identifies technical formations and statistical signals.

The Strategy Explainer Agent translates technical findings into clear, human-readable explanations.

The Risk Critic Agent evaluates uncertainty, false signals, and risk exposure.

The Coordinator Agent manages workflow and aggregates outputs into a final response.

This modular architecture ensures that each stage can be evaluated independently, supporting verification and iterative improvement.

Tools and APIs

The system will use a large language model API, Python financial analysis libraries, CSV file upload and parsing tools, optional historical market datasets, and a user-facing interface built with Streamlit. The system is designed for educational analysis only and will not connect to live brokerage accounts or execute trades.

Feasibility and Risks

Potential challenges include inaccurate pattern detection, noisy financial data, hallucinated explanations, and overconfidence in predictions. Constraints include API rate limits, dataset limitations, and the semester timeline.

Mitigation strategies include rule-based validation checks, confidence scoring, and requiring the critic agent to verify outputs before presenting them to users.

Timeline

Weeks 1–2: Proposal and architecture design

Weeks 3–4: Front-end prototype

Weeks 5–6: Multi-agent implementation

Week 7: Data integration and testing

Week 8: Evaluation experiments

Week 9: Refinement

Week 10: Final report and presentation

Expected Outcome

The final system will demonstrate how a multi-agent AI architecture can interpret financial chart data and provide educational technical analysis. Evaluation will measure clarity of explanations, correctness of detected patterns, and perceived usefulness to beginner traders. The project aims to show that agentic AI can improve financial literacy without performing automated trading.